Lesson plan · for the teacher

Logging data you can trust

IoT · Level 3 · Lesson 1

Time
40 min

Materials
A thermometer: a cheap dial or digital one is fine; A ruled log sheet and a pen by the door; Chalk or markers for sorting the readings; Optional: a phone with the free phyphox app, or a micro:bit (it has a temperature sensor)

Learning goals

- Plan a data log: what to measure, where, how often, with a time and unit on every reading.
- Check a sensor against a reference before trusting it.
- Clean glitches openly, and use the median when a few bad readings would drag the average.

Activity: A paper log on a rota

1. Warm-up (5 min): Write “Classroom: 85 °C at 7 am” on the board. Ask: could this be true, and what could cause it? Collect ideas, then explain the sensor that reads 85 °C as it powers up.
2. Plan (8 min): Agree one question with the class, such as “Is our room too hot after lunch?” Decide together what to measure, where to hang the thermometer (desk height, in the shade) and how often to read it. Rule a log sheet with columns for date, time, reading, unit and initials.
3. Check (7 min): Put two thermometers side by side for a few minutes. Pupils read both and write the readings in two columns. Is the difference steady? Explain an offset, and why we check before we trust.
4. Log (12 min): Set the rota so each pupil has an hour this week. Practise three readings now: eyes level with the scale, then the time, the number and the unit. Show how to write “missed” instead of leaving a blank.
5. Wrap-up (8 min): Give a sample afternoon: 31, 29, 85, 30, 28. Pupils sort it, find the median (30), work out the average with and without the 85, and write a one-line cleaning note.

Before class, hang the thermometer in the shade at desk height and check that pupils can read it. Use a dial or digital thermometer with young pupils; if a glass one breaks, an adult clears it up. If you use a sensor board, run it on batteries only.

Discussion

- Where else at school could a glitch mislead a decision?
- When should a surprising reading stay in the data?
- Who should be able to check how the data was cleaned?

Exit ticket

- Readings: 30, 28, 85, 31, 29 °C. What is the median, and what goes in the cleaning note?

Try it in class tomorrow
A paper log on a rota: Agree a question with your class, such as “Is our room too hot after lunch?” Hang a thermometer in the shade at desk height and set a rota: each hour, one pupil writes the

time, the reading and the unit on a ruled sheet by the door. If you can, add a phone app such as phyphox, or a sensor board, beside it. At the end of the week, pupils sort the afternoon readings, find the median, mark any glitch and write a cleaning note.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Zainab's pupils read the garden's soil moisture at noon each day: 62, 60, 0, 58, 0, 57 and 55 %. The two zeros came from a loose wire. Before any cleaning, which summary gives the fairest typical value?**

◯ The average ◯ The median ◯ The lowest reading

2 **A new sensor that always reads exactly 1 °C higher than a trusted thermometer is useless.**

◯ True ◯ False

3 **Put the steps of a trustworthy data log in order.**

☐ Summarise with a chart and the median

☐ Copy the log, leave out glitches and note why

☐ Log every reading with its time, unit and place

☐ Check the sensor beside a reference

☐ Decide what to measure, where and how often

☐ Write down the question you want answered

4 **In the Data logger lab, with "Needs water below" at 40 %, you leave out the four glitches (two readings of 0 and two of 100). Which numbers change a lot?**

☐ The average ☐ The lowest reading ☐ The median ☐ The hours too dry

☐ The highest reading

5 **Five afternoon readings came in: 31, 29, 85, 30 and 28 °C. Work out the summaries.**

Sorted, the middle reading is _____ °C, so that is the median. Leaving out the 85, the average of the other four is _____ °C.

Word bank: 30 · 29.5 · 40.6 · 85

✎ **Which question about your school could your pupils answer with a week of readings? What would they measure, where, and how often?**